



FACULTÉ
D'INFORMATIQUE

CHAPTER 2 PART 2

RELATIONAL DATABASE – IMPLEMENTING WITH SQL

Zakia Challal

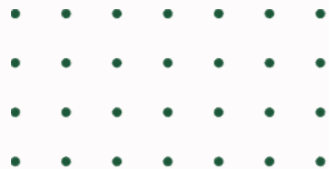
zakiachallal@gmail.com

Computer Science Engineering. 2nd year. 2024-2025



Content

- 01** SQL
- 02** Create tables
- 03** Integrity constraints
- 04** Insert data
- 05** Update data
- 06** Delete data
- 07** Exercise



Introduction

- In the previous part of this chapter we have learn how to design a database and lead to normalized set of relations.
- In this part of the chapter we will see how to implement these relations using a DBMS and the Structured Query Language (SQL)

SQL

- **SQL (Structured Query Language)** is a standard language used to interact with relational databases. It allows users to **store, retrieve, modify, and manage** data efficiently.
- **Main Uses of SQL:**
 - Querying Data** – SELECT
 - Inserting Data** – INSERT INTO
 - Updating Data** – UPDATE
 - Deleting Data** – DELETE
 - Creating & Modifying Database Structures** – CREATE TABLE, ALTER TABLE, DROP TABLE

Course example

Student

StudID	First name	Last name	Birthday	Address	email	DepID

Department

DepID	Name	Head	Budget

Create Table

- **Syntax:**

Create table Table_Name (attribute1 type, attribute2 type,...);

- **Example:**

Create table Student (StudID integer, FirstName varchar(20), LastName varchar(20), Birthday date, email varchar(50), DepID integer);

Create table Department (DepID integer, Name varchar(50), Head varchar(40));

- **Note:** SQL is not case sensitive.

Data Type

● Character data type

- **Char(n)**: Fixed-length character string (max 2000 bytes)
- **Varchar(n)**: Variable-length character string (max 4000 bytes).

● Numeric data type

- **Number(p,s)**: fixed and floating-point numbers (p:Total number of digits, s: Number of digits after the decimal)
- **Integer/int**: equivalent to Number (38,0)

● Date time data type

- **Date**: Stores date and time (YYYY-MM-DD HH:MI:SS).
- **Timestamp**: Stores date and time with fractional seconds (n).

● Large Object data type

- **Clob**: Stores large text data.
- **Blob**: Stores large binary data (e.g., images, videos)

Integrity constraint

Reminder

- An **integrity constraint** is a set of rules that ensure the coherence of data in a relational database. These constraints prevent invalid data entry and maintain the logical correctness of relationships between tables.

Constraint Type	Purpose
Domain Constraint	Restricts values based on data type and rules
Entity Integrity	Ensures every table has a unique identifier (Primary key)
Referential Integrity	Maintains valid relationships between tables (Foreign key)
Unique Constraint	Ensures uniqueness of certain attributes
NOT NULL Constraint	Prevents missing (NULL) values
CHECK Constraint	Imposes additional rules on column values

Integrity constraint

Primary Key

Different ways to define the primary key:

When creating table:

- Create table Student (StudID integer **Primary key**, FirstName

Or

- Create table Student (StudID integer, FirstName varchar(20), LastName varchar(20), Birthday date, email varchar(50), DepID integer, **Constraint PK1 Primary key (StudID)**);

Or After creating table

- **Alter table** Student **add constraint PK1 Primary key (StudID);**

Integrity constraint

Foreign Key

Different ways to define the Foreign key:

When creating table:

- Create table Student (StudID integer Primary key, FirstName, DeptID integer **Foreign key references Department(DeptID)**);

Or

- Create table Student (StudID integer Primary key, FirstName, DeptID integer, **Constraint FK1 Foreign key (DeptID) references Department(DeptID)**);

Or After creating table

- **Alter table** Student **add constraint FK1 Foreign key (DeptID) references Department(DeptID)**;

Integrity constraint

Foreign Key- ON DELETE CASCADE/ ON DELETE SET NULL

- We can't delete tuples from the parent table if related tuples exist in the child table.
Example, we can't delete a department if there are Student affected to that department.
We can force deletion by adding : **ON DELETE CASCADE** or **ON DELETE SET NULL** to the constraint definition.
- **ON DELETE CASCADE:** delete tuples in the child table automatically.
Ex: delete student of a department when the department is deleted
- **ON DELETE SET NULL:** replace the value of the foreign key with Null.
Ex: replace the value of the DeptID in the student table with Null value.
- **Example:** Create table Student (StudID integer Primary key, FirstName, DeptID integer
Foreign key references Department(DeptID) ON DELETE CASCADE);

Integrity constraint

Unique

Different ways to define the constraint **UNIQUE**:

When creating table:

- Create table Student (StudID integer Primary key, FirstName, email varchar(50) **Unique**)

Or

- Create table Student (StudID integer, FirstName varchar(20), LastName varchar(20), Birthday date, email varchar(50), DepID integer, **Constraint U1 Unique(email)**);

Or After creating table

- **Alter table Student add constraint U1 Unique(StudID);**

Integrity constraint

Not Null

Different ways to define the constraint **NOT NULL**:

When creating table:

- Create table Student (StudID integer Primary key, FirstName varchar(20) **Not Null**,.....)

Or

- Create table Student (StudID integer, FirstName varchar(20), LastName varchar(20), Birthday date, email varchar(50), DepID integer, **Constraint C1 Check (FirstName is not null)**);

Or After creating table

- **Alter table** Student **add constraint C1 Check (FirstName is not null);**

Integrity constraint

Check

Different ways to define the constraint **CHECK**:

When creating table:

- Create table Student (StudID integer Primary key,....., Birthday date **Check (Birthday < Sysdate)...**);
- Or
- Create table Student (StudID integer, FirstName varchar(20), LastName varchar(20), Birthday date, email varchar(50), DepID integer, **Constraint C1 Check (Birthday < Sysdate)**);

Or After creating table

- **Alter table** Student **add constraint C1 Check (Birthday < Sysdate);**

Integrity constraint

Default

Different ways to define the constraint **DEFAULT**:

When creating table:

- Create table Department(DeptID integer Primary key, Name varchar(20), Budget int **Default 0**);

Or After creating table

- **Alter table** Department **Modify** Budget **Default 0**;

Alter table

Add column

- Add one column

Alter table Student **Add** Phone varchar(15);

- Add multiple column

Alter table Department **Add** (Capacity int, CreationDate Date);

Alter table

Drop column

- Drop one column

Alter table Student **Drop column** Phone;

- Drop multiple column

Alter table Department **Drop** (Capacity , CreationDate);

Alter table

Modify column

- **Modify data type**

Alter table student **Modify** FirstName **varchar(50);**

- **Modify not null/null constraint**

Alter table Student **modify** LastName **Not Null;**

Alter table Student **modify** LastName **Null;** (correct only if a not null constraint is already defined)

- **Add Default Value**

Alter table Department **Modify** Budget **Default 0;**

Alter table

Rename column

- **ALTER TABLE** table_name **RENAME COLUMN** old_coloumn **TO** new_column;

Ex: **Alter table** Department **Rename Column** Head **To** HeadDept;

Rename table

- **RENAME** old_table **TO** new_table;

Ex: **Rename** Department **to** Dept;

Drop table

- **DROP TABLE** nom_table;

Ex: **Drop Table** Dept;

Insert data into a table

- **INSERT INTO** table_name (column1, column2, ...) **VALUES** (value1, value2, ...);

Ex: insert into dept (DeptID, Name, HeadDept,Budget) values (1, 'AI&SD', 'Djidel Faiza', 0);

- If we insert values **for all columns in order**, we can omit the column list:
insert into dept values (1, 'AI&SD', 'Djidel Faiza', 0);

Delete data from a table

- **DELETE FROM** table_name **WHERE** condition;

Example:

- Delete From Student Where StudID='1234';

Delete From Student;

Update data in a table

- **UPDATE** table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
- Ex:
Update Student
Set email='zakiachallal@gmail.com'
Where StudID= 1234;



The END